

Supplementary Information: Payload-Aware Autonomous ADC Linker Design

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1 Full context ranking (all 38 contexts)

Handle	Trigger	Payload	mean	best	SD	seeds
Maleimide	Non-cleavable	immunomodulator	0.85	0.85	0.01	2
Oxyamine	Non-cleavable	cytotoxin	0.84	0.85	0.01	2
Maleimide	Non-cleavable	cytotoxin	0.83	0.84	0.01	2
Maleimide	Non-cleavable-rigid	immunomodulator	0.83	0.84	0.00	2
Bromoacetamide	Non-cleavable	cytotoxin	0.83	0.83	0.00	2
Maleimide	Non-cleavable-rigid	cytotoxin	0.82	0.82	0.00	2
Tetrazine	Non-cleavable	cytotoxin	0.82	0.82	0.00	2
NHS_ester	Non-cleavable	cytotoxin	0.81	0.81	0.00	2
Maleimide	Non-cleavable	oligonucleotide	0.78	0.79	0.01	2
Maleimide	Non-cleavable-rigid	oligonucleotide	0.76	0.76	0.01	2
Disulfide	Non-cleavable	cytotoxin	0.74	0.76	0.02	2
DBCO	Non-cleavable	immunomodulator	0.72	0.75	0.04	2
DBCO	Non-cleavable	cytotoxin	0.70	0.73	0.04	2
Maleimide	Val-Cit-PABC	immunomodulator	0.72	0.72	0.00	2
DBCO	Non-cleavable	oligonucleotide	0.70	0.71	0.00	2
Maleimide	Val-Cit-PABC	cytotoxin	0.70	0.71	0.00	2
TCO	Non-cleavable	cytotoxin	0.69	0.69	0.01	2
DBCO	Non-cleavable-rigid	immunomodulator	0.68	0.69	0.01	2
Bromoacetamide	Val-Cit-PABC	cytotoxin	0.66	0.68	0.02	2
DBCO	Non-cleavable-rigid	cytotoxin	0.66	0.67	0.01	2
Oxyamine	Val-Cit-PABC	cytotoxin	0.67	0.67	0.00	2
NHS_ester	Val-Cit-PABC	cytotoxin	0.65	0.67	0.02	2
DBCO	Non-cleavable-rigid	oligonucleotide	0.66	0.67	0.00	2
Tetrazine	Val-Cit-PABC	cytotoxin	0.64	0.67	0.02	2
Disulfide	Val-Cit-PABC	cytotoxin	0.61	0.63	0.02	2
TCO	Val-Cit-PABC	cytotoxin	0.58	0.59	0.01	2
DBCO	Val-Cit-PABC	immunomodulator	0.55	0.56	0.01	2
DBCO	Val-Cit-PABC	cytotoxin	0.53	0.53	0.01	2
Maleimide	Val-Cit-PABC	oligonucleotide	0.45	0.45	0.00	2
DBCO	Val-Cit-PABC	oligonucleotide	0.38	0.40	0.02	2
Bromoacetamide	Glucuronide	cytotoxin	0.08	0.08	0.00	2
Oxyamine	Glucuronide	cytotoxin	0.08	0.08	0.00	2
Tetrazine	Glucuronide	cytotoxin	0.08	0.08	0.00	2
Maleimide	Glucuronide	cytotoxin	0.08	0.08	0.00	2
NHS_ester	Glucuronide	cytotoxin	0.08	0.08	0.00	2
Disulfide	Glucuronide	cytotoxin	0.08	0.08	0.00	2
TCO	Glucuronide	cytotoxin	0.07	0.07	0.00	2
DBCO	Glucuronide	cytotoxin	0.06	0.06	0.00	2

2 Payload-class re-scoring (mean composite by payload rule)

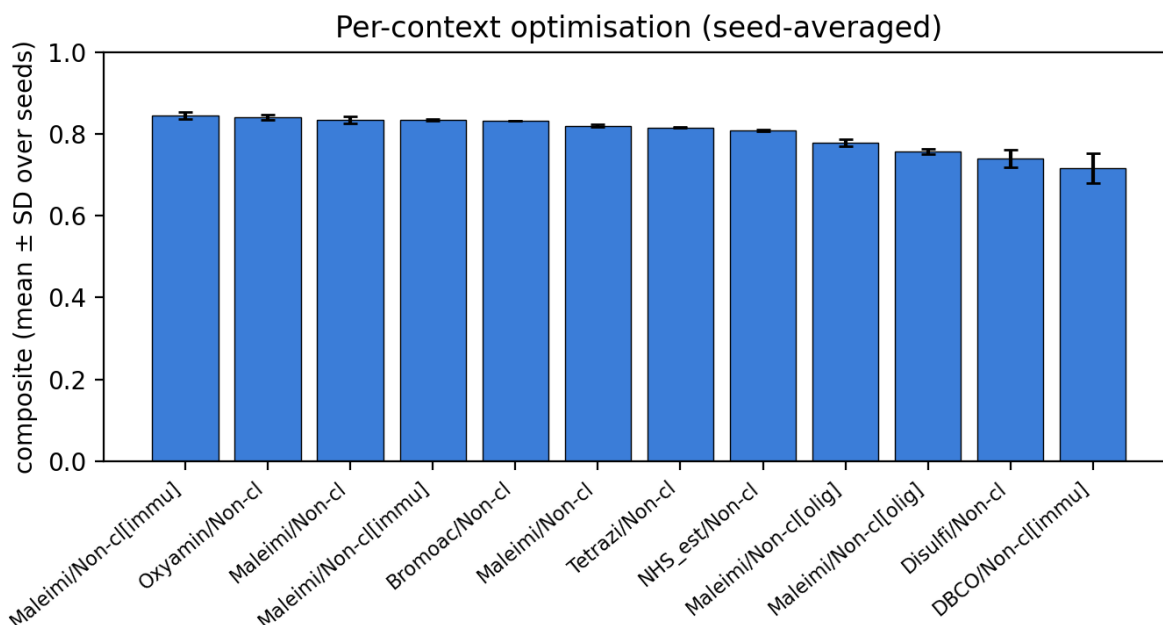


Figure 1: Per-context optimisation, mean $\pm$ SD over seeds (top contexts). Modest seed variance supports the ranking.

Handle	Trigger	cytotoxin	oligonucleotide	immunomodulator
Maleimide	Non-cleavable	0.83	0.78	0.85
Maleimide	Non-cleavable-rigid	0.82	0.76	0.83
DBCO	Non-cleavable	0.70	0.70	0.72
Maleimide	Val-Cit-PABC	0.70	0.45	0.72
DBCO	Non-cleavable-rigid	0.66	0.66	0.68
DBCO	Val-Cit-PABC	0.53	0.38	0.55

3 Table S1: Stability-surrogate calibration

Mechanistic class	stability	expected
acid-labile hydrazone	0.00	low
redox disulfide	1.00	intermediate
protease dipeptide	1.00	high
non-cleavable thioether	1.00	high

4 Commercial linker benchmark

Name	ADC / class	composite	QED	MW
mc-Val-Cit-PABC	brentuximab vedotin (Ad...	0.52	0.116	573
mc-Val-Ala-PABC	Val-Ala class	0.59	0.244	487
mc-GGFG	trastuzumab deruxtecan ...	0.58	0.128	530
MCC (non-cleavable)	trastuzumab emtansine (...)	0.67	0.737	237
SPDB disulfide	mirvetuximab soravtansi...	0.74	0.600	229
sulfo-SPDB	hindered disulfide class	0.76	0.444	309
AcBut hydrazone	gemtuzumab ozogamicin (...)	0.00	0.340	236
beta-glucuronide	glucuronide linker class	0.00	0.156	508
maleimidocaproyl (mc)	mc handle	0.68	0.514	211

5 Table S2: Top designed linkers (full SMILES)

Context	Payload	Score	SMILES
Bromoa/Non-cl	cytot	0.83	<chem>O=C(CBr)NC(=O)Cn1nc(NC(=O)NCc2ccccc2)c(=O)[nH]c1=O</chem>
Oxyami/Non-cl	cytot	0.83	<chem>NOCC(=O)Nc1c(C(=O)NCc2ccccc2)[nH][nH]c1=O</chem>
Oxyami/Non-cl	cytot	0.83	<chem>NOc1c(N)nc(N2CCN(C(=O)NCc3ccccc3)CC2)cc1=O</chem>
Maleim/Non-cl	immun	0.83	<chem>O=C(NCc1ccccc1)NS(=O)(=O)c1ccc(S(=O)(=O)N2C(=O)C=CC2=O)cc1</chem>
Oxyami/Non-cl	cytot	0.82	<chem>NOCCn1c(=O)c(C(=O)NCc2ccccc2)cc2cnc(N)nc21</chem>
Bromoa/Non-cl	cytot	0.82	<chem>O=C(Cn1cnc2c(nnn2CC(=O)NC(=O)CBr)c1=O)NCc1ccccc1</chem>
Oxyami/Non-cl	cytot	0.82	<chem>NOCCC(=O)c1c(C(=O)NCc2ccccc2)[nH]c(=O)[nH]c1=O</chem>
Tetraz/Non-cl	cytot	0.82	<chem>Cc1nnc(NS(=O)(=O)CCNC(=O)NCc2ccccc2)nn1</chem>

6 Table S3: Boltz-2 co-folding with cathepsin B (full panel)

Linker	type	ipTM	pLDDT	Kd (nM)
Maleimide/Val-Cit-PABC	designed	0.85	87.1	6.4
Bromoacetamide/Val-Cit-PABC	designed	0.77	83.8	14.1
Oxyamine/Val-Cit-PABC	designed	0.78	86.1	75.8
NHS_ester/Val-Cit-PABC	designed	0.89	86.8	38.1
Tetrazine/Val-Cit-PABC	designed	0.84	83.5	30.8
Disulfide/Val-Cit-PABC	designed	0.80	82.8	7.4
Maleimide/Non-cleavable-r...	negative	0.86	87.9	787.6
mc-Val-Cit-PABC	commercial	0.84	86.0	43.1
mc-Val-Ala-PABC	commercial	0.64	84.9	185.6
maleimidocaproyl (mc)	commercial	0.83	91.7	113836.3
AcBut hydrazone	commercial	0.91	91.1	16158.0

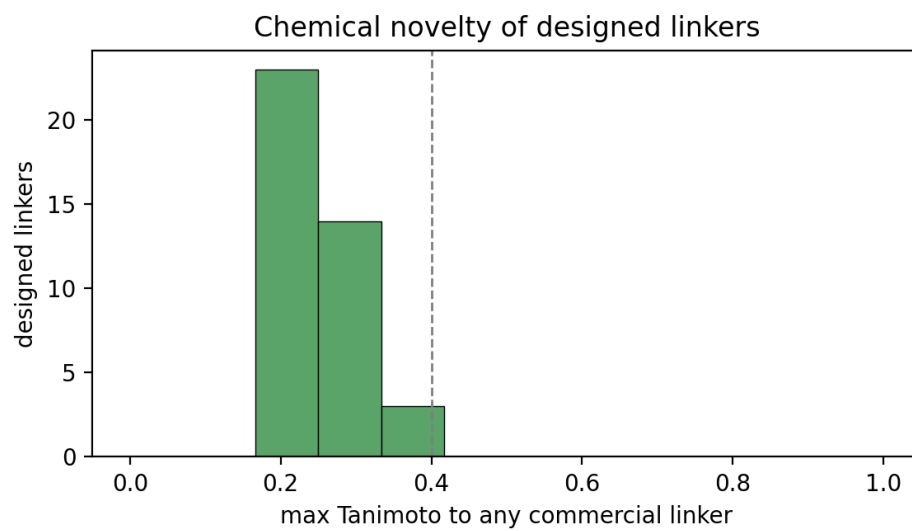


Figure 2: Chemical novelty: nearest-neighbour Tanimoto of each designed linker to any commercial linker. Most designs are distinct chemotypes (similarity < 0.4), not rediscoveries.